

TEODORA BEČEJAC

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EDUCATION

Faculty of Technical Sciences, University of Novi Sad

Sept 2021 – present

B.Sc. Computing and Control Engineering | Teaching Assistant, Computer Architecture (Sept 2024 - July 2025)

TECHNICAL SKILLS

Languages: Java, Go, C#, JavaScript/TypeScript, Python, C++, SQL, Kotlin

Frameworks: Spring Boot 3, .NET 7, Node.js/Express, Angular, Vue.js, Android SDK, WPF, React

Distributed Systems: Microservices, SAGA/Event-Driven Architecture, gRPC, RabbitMQ (AMQP), Docker, WebSockets, OAuth2/JWT

Data & Algorithms: PostgreSQL + PostGIS, Neo4j, MongoDB, Oracle 19c, SQLite, Firebase, A*/CBS, Genetic Algorithms, OpenCV

Observability & Sec.: Prometheus, Grafana, Jaeger, Loki (LGTM), OpenTelemetry, OWASP, Bouncy Castle, X.509/PKCS#12

Practices: DDD, TDD, MVVM, Hexagonal Architecture, GitFlow, Docker Compose, Scrum/Jira

SELECTED PROJECTS

AeroMesh: Multi-Agent UAV Fleet Routing System

2026

MathWorks Excellence in Innovation Challenge Submission

MATLAB/Simulink · C++17 · Go · React · Docker · RabbitMQ

- Designed a full-stack 4-layer **Software-in-the-Loop (SIL)** co-simulation: MATLAB/Simulink physics & sensor twin → C++17 real-time motion planner → Go/GA logistics engine → React dashboard with a **custom mission DSL** — replicating the architecture of a commercial drone fleet management platform.
- A* on a 3D voxel grid + **Conflict-Based Search (CBS)** with 3 original edge-case extensions; heuristics encode live Perlin-noise **wind cost** and inter-drone **conflict penalties** for deadlock-free routing in **4D space (XYZ + Time)**.
- Island-Model GA** (3 islands, 200 gen) solved the **NP-Hard VRP**: **19.1% avg energy saving** over greedy (peak **20.9%**), near-linear scalability (64 ms → 223 ms, 3.5× factor), and quantified payload impact (**+89.5%** energy, heavy vs light); **Chaos Monkey** validated production-grade resilience under server crashes and in-flight drone failures.

Wanderlust: Polyglot Microservices Tourism Ecosystem

2025

Production-Grade Distributed Systems

Go · Node.js · Neo4j · MongoDB · PostgreSQL · RabbitMQ · Docker · LGTM

- Architected **15+ containerised microservices** with **Polyglot Persistence**: Neo4j for **O(1) social-graph traversals**, MongoDB for high-write telemetry, PostgreSQL for **ACID-compliant** financial ledgers — the database-per-domain strategy used at Netflix and Uber.
- Orchestration-based SAGA** eliminates 2-Phase Commit: Stakeholder Service acts as state-machine orchestrator across all 3 DBs with **compensating transactions** guaranteeing consistent state on any partial failure; full **LGTM** stack (**OpenTelemetry** tracing, Prometheus/Grafana, Loki/LogQL) delivers Spotify-grade production observability.

entinel PKI: Zero-Knowledge Trust Infrastructure

2025

Enterprise Cryptography & Security

Spring Boot 3 · Java 17 · Bouncy Castle · PostgreSQL · OAuth2

- X.509-compliant Certificate Authority** (Root → Intermediate → End-Entity) with **cascading revocation**: a single revocation atomically propagates through the entire descendant subtree and emits **signed CRLs** — the trust-chain model behind every HTTPS connection on the internet.
- Zero-Knowledge Credential Vault** via **Hybrid Encryption** (ephemeral AES-256 + RSA/EC key wrapping): server stores only ciphertext and is **mathematically incapable** of recovering any secret; **OWASP-hardened** with **JWT session fingerprinting**, ASN.1 sanitisation, and PKCS#12 encapsulation under Master AES-256 (**HSM-level** in software).

Rabbit: High-Concurrency Social Platform — Custom Infrastructure

2025

Distributed Middleware Engineering

Java Spring Boot · Angular · PostgreSQL · WebSockets · Prometheus · Grafana

- Engineered a **proprietary Dual-Mode Message Queue** from scratch: AMQP-style **Direct Exchange** (vet-location ingestion) and **Fanout Exchange** (ad broadcasting), runtime-swappable via Spring Profiles — a level of **broker internals** mastery rarely seen outside core platform teams.
- Custom Load Balancer** (Round Robin / Least-Connections, active health-check probing, **zero-downtime failover** mirroring NGINX); **O(1) sliding-window Rate Limiter** at 5 req/min/IP; **Bloom Filters** for sub-ms uniqueness checks; **pessimistic row-level locking** eliminates **lost-update race conditions** under burst concurrency.

Travel Tales: Geospatial Tourism Platform (16-Engineer Team)

2024–2025

.NET 7 · Angular · PostgreSQL + PostGIS · xUnit TDD · Jira/Scrum

Delivered **4 Domain-Driven bounded contexts** as one of 16 engineers (strict GitFlow, mandatory code review, 2-week Scrum sprints); built "**Gavrilo**" — a context-aware AI chatbot with **TTS-narrated** tour guidance and quiz-driven **behavioural personalisation** — serving as the platform's primary engagement layer.

- PostGIS** spatial queries drive sub-second real-time encounter triggers across thousands of concurrent checkpoints; **anti-cheat validation** enforces physical presence (>35% path completion, time-delta realism) robust against **GPS spoofing**; **Best-Price** engine resolves overlapping Bundle/Coupon/Flash Sale strategies in a single pass.

AWARDS & ACHIEVEMENTS

- Recipient of **30+ diplomas** in regional and national Mathematics competitions.
- Teaching Assistant**, Computer Architecture: Assembly language and low-level optimisation, mentored undergraduates (Sept 2024 – July 2025).
- Youth Mathematics Mentor and former President of the Ecology Club (2 years).